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TECHNOLOGICAL UNIVERSITY DUBLIN

CITY CAMPUS

DT249 – BSc. (Honours) in Information Systems and Information Technology
DT255 – BSc. (Honours) in Information Systems and Information Technology

Year 4

SEMESTER 2 EXAMINATIONS 2021/22

Machine Learning for Data Analytics

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Duration: 2 hours

Instructions

Answer ALL questions.
Question 1 carries 40 marks and questions 2 and 3 carry 30 marks each.

1. (a) What is **supervised machine learning?** Give an example!

(5 marks)

(b) Explain on an **example** what can go wrong when a machine learning classifier uses the wrong **inductive bias**.

(5 marks)

(c) Table 1 shows predictions made for a categorical target feature by a model for a test dataset. Based on this test set, calculate the evaluation measures listed below and **explain exactly** what they would tell us with regard to a **spam detection** dataset (assume that false is 'ham' and true is 'spam').

(i) A **confusion matrix**

(6 marks)

(ii) The **misclassification rate**

$$misclassification\ rate = \frac{(FP + FN)}{(TP + TN + FP + FN)}$$

(4 marks)

(iii) The **precision, recall** and **F1 measure**

$$precision = \frac{TP}{(TP + FP)}$$

$$recall = \frac{TP}{(TP + FN)}$$

$$F_1\ measure = 2 \times \frac{(precision \times recall)}{(precision + recall)}$$

(12 marks)

(iv) The **average class accuracy (harmonic mean)** (round to 3 decimal places).

(8 marks)

$$average\ class\ accuracy_{HM} = \frac{1}{\frac{1}{|levels(t)|} \sum_{l \in levels(t)} \frac{1}{recall_l}}$$

Table 1: The predictions made by a model for a categorical target on a test set of 20 instances

ID	Target	Prediction	ID	Target	Prediction
1	false	false	11	false	false
2	true	true	12	false	false
3	false	false	13	false	false
4	true	true	14	false	false
5	false	false	15	true	true
6	false	false	16	false	false
7	true	false	17	true	true
8	true	true	18	true	true
9	true	true	19	false	false
10	true	true	20	false	false

2 (a) A data analyst building a k-nearest neighbour model for a continuous prediction problem is considering appropriate values to use for k on an *imbalanced training set*.

(i) Initially the analyst uses a simple average of the target variables for the k nearest neighbours in order to make a new prediction. After experimenting with small values for k in the range $0 - 5$ it occurs to the analyst that they might get very good results if they keep increasing k to a value closer to the total number of instances in the training set. Do you think the analyst is likely to get good results using these values for k?

(5 marks)

(ii) If the analyst was using a distance weighted averaging function rather than a simple average for their predictions would this have made their idea any more useful?

(5 marks)

(iii) By using a different distance metric than the standard Euclidean Distance, would any of the previous answers change? Provide an explanation to your answer.

(5 marks)

(b) Table 2 on the next page lists a sample of data from a census. There are four descriptive features in this dataset (AGE, EDUCATION, MARITAL STATUS, OCCUPATION) and the target feature ANNUAL INCOME has 3 levels (<25K, 25K–50K, >50K). Note, Table 3, also on the next page, lists some equations that you may find useful for this question.

(i) Calculate the ENTROPY for this dataset.

(5 marks)

(ii) When building a decision tree, we must partition the data into homogeneous subsets. What is the metric used to decide on the partitions? How it relates to the entropy of the dataset?

(5 marks)

(iii) In the case that we have a continuous descriptive feature, what is the procedure to create the partitions using information gain?

(5 marks)

Table 2: Census data for the ID3 Algorithm Question

ID	AGE	EDUCATION	MARITAL STATUS	OCCUPATION	ANNUAL INCOME
1	39	bachelors	never married	transport	$25K$–$50K$
2	50	bachelors	married	professional	$25K$–$50K$
3	18	high school	never married	agriculture	$<25K$
4	28	bachelors	married	professional	$25K$–$50K$
5	37	high school	married	agriculture	$25K$–$50K$
6	24	high school	never married	armed forces	$<25K$
7	52	high school	divorced	transport	$25K$–$50K$
8	40	doctorate	married	professional	$>50K$

Table 3: Equations from information theory.

$$H(\mathbf{f}, \mathcal{D}) = -\sum_{l \in levels(f)} P(f = l) \times log_2(P(f = l))$$

$$rem(\mathbf{f}, \mathcal{D}) = \sum_{l \in levels(f)} \frac{|\mathcal{D}_{f=l}|}{|\mathcal{D}|} \times H(t, \mathcal{D}_{f=l})$$

$$IG(\mathbf{d}, \mathcal{D}) = H(\mathbf{t}, \mathcal{D}) - rem(\mathbf{d}, \mathcal{D})$$

3. Table 4 lists a dataset of the previous decision made by a couple regarding whether or not they would wait for a table at a restaurant (i.e., the feature WAITED is the target feature in this domain).

(a) Calculate the probabilities (to four places of decimal) that a **naïve Bayes** classifier would use to represent this domain.

(18 marks)

(b) Assuming conditional independence between features given the target feature value, calculate the **probability** of each outcome (WAITED=Yes, and WAITED=No) for the following restaurant for this couple (marks will be deducted if workings are not shown, round your results to four places of decimal)

BAR=False, PATRONS=None, PRICE=Expensive

(10 marks)

(c) What prediction would a **naïve Bayes** classifier return for the above restaurant? **Explain** what the result means and how to interpret probabilistic predictions.

(2 marks)

Table 4: A dataset describing the previous decisions made by an individual about whether to wait for a table at a restaurant.

ID	BAR	PATRONS	PRICE	WAITED
1	False	Some	Expensive	Yes
2	False	Full	Cheap	No
3	True	Some	Cheap	Yes
4	False	Full	Cheap	Yes
5	False	Full	Expensive	No
6	True	Some	Reasonable	No
7	True	None	Cheap	No
8	False	Some	Reasonable	No
9	True	Full	Cheap	No
10	True	None	Reasonable	Yes